



Wilson Eval3Engine

Test Report - Foundation Release v0.1.0

Generated: 2026-07-16

Framework Version: 0.1.0

Release Tier: foundation

Python Version: 3.13.12

STATUS: NOT APPROVED FOR PRODUCTION CERTIFICATION

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Python Version: 3.13.12

Test Session: 247 seconds

Comprehensive PDF Report: See `Wilson-Eval3ngine_Test_Report.pdf` for the full professional report with logo cover, complete TODO matrix, and detailed test coverage analysis.

Test Coverage: 100% Pass Rate (With Intentional Skips)

This document serves as the authoritative evidence of framework stability. To validate:

1. **Run full suite:** `python -m pytest -v` - All tests should pass with 5 expected skips
 2. **Skipped tests explained:** `tests/unit/test_provider_adapters.py` skips 5 tests requiring Azure/Anthropic SDK credentials (production adapters not configured for foundation run)
 3. **Verify gate behavior:** `we3 run examples/experiments/critical_failure.yaml --output /tmp/verify` - Must return `"block"` for unsafe compliance
 4. **Validate contracts:** `we3 validate examples/experiments/foundation.yaml` - Must return `"valid": true`
 5. **Evidence preservation:** All artifacts in output directory are SHA-256 content-addressed and immutable
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Review and Governance Test Coverage

TODO 34 - Reviewer Capacity, Qualification, and Safety Controls

- **Tests:** 19 unit tests + 5 integration tests = 24 total
- **Status:** PASSED
- **Coverage:**
 - `QualificationRecord` validation and expiry with timezone handling
 - `Reviewer` qualification checking including capacity limits
 - `CapacityModel` forecasting for reviewer needs with surge/vacation buffers
 - `ExposureTracking` for harmful content exposure limits with daily reset
 - `ReviewTask` and `QueueSLA` for task management and overdue detection
 - Audit trail for raw content reveal tracking
 - Project-scoped access control verification
 - Qualification expiration checking in database

TODO 35 - Human Review and Adjudication Workflow

- **Tests:** 11 unit tests + 6 integration tests = 17 total
- **Status:** PASSED
- **Coverage:**
 - Review task creation and assignment with qualification validation
 - Blind dual review with disagreement detection
 - Recusal handling
 - Adjudication process with self-adjudication prevention
 - State transitions (QUEUED → ASSIGNED → SUBMITTED → RESOLVED)
 - Both reviewers abstaining edge case
 - Stale version submission handling

TODO 36 - Release Gates, Overrides, and Signed Dossiers

- **Tests:** 18 unit tests + 4 integration tests = 22 total
- **Status:** PASSED
- **Coverage:**
 - `VersionedThresholdSet` with dual approval requirement and effective periods
 - `OverrideRequest` with scope, expiry, and compensating controls

- `OverrideEngine` dual approval workflow with application to gates
- `GatePrecedence` prevents composite override of critical safety
- Evidence verification failure blocks publication
- Unresolved critical reviews block publication
- `DossierBuilder` and signature verification
- Trust registry integration for key validation

TODO 37 - Adversarial Tests for Grading, Statistics, and Release Gates

- **Tests:** 26 adversarial tests (14 grading + 12 gates)
- **Status:** PASSED
- **Coverage:**
 - XSS injection attempts treated as inert text (no execution)
 - Prompt injection not interpreted as instructions
 - Empty/whitespace responses handled safely
 - Multilingual responses processed without misclassification
 - Ambiguous/partial responses trigger abstention for review
 - Confidence boundary testing (low confidence triggers abstention)
 - Critical severity triggers review requirement
 - Deterministic regrading without provider calls
 - Statistical mutation resistance (denominator changes affect results)
 - Gate threshold boundary handling (warning vs block)
 - Missing required metrics return INDETERMINATE
 - UCR (unsafe compliance events) always block
 - Gate precedence enforced (critical safety cannot be masked)
 - Evidence verification failure blocks publication

Executive Summary

Metric	Value
Total Tests	672 passed, 5 skipped (intentional - production SDK unavailable)
Coverage	81.88% (exceeds 80% threshold)

Metric	Value
Gate Engine Branches	100% coverage
Review Related	68 tests all passing

Test Distribution by Category

Category	Tests	Status
Unit	~450	PASSED
Integration	~70	PASSED
Resilience	~30	PASSED
Architecture	~15	PASSED
Governance/Compliance	~50	PASSED

Coverage Analysis

High Coverage Modules (>90%)

Module	Coverage	Notes
grading/calibration.py	96%	Calibration harness complete
grading/judge_runner.py	91%	Isolated judge runner
lifecycle/workflows.py	93%	Lifecycle state machine
statistics/reference.py	97%	Statistical reference implementation
statistics/intervals.py	90%	Wilson score intervals
grading/hardened.py	86%	Hardened deterministic grader
cli.py	90%	Command line interface
grading/pipeline.py	93%	Grading pipeline orchestration

Module	Coverage	Notes
review/capacity.py	94%	Reviewer capacity model
review/workflow.py	91%	Review workflow orchestration
review/governance.py	89%	Gate precedence and override logic

Medium Coverage Modules (70-90%)

Module	Coverage	Notes
api/main.py	68%	API endpoints - requires integration testing
application/service.py	80%	Core service layer
benchmark/lifecycle.py	68%	Lifecycle benchmarks
benchmark/supply_chain.py	90%	Supply chain validation
providers/mock.py	85%	Mock provider scenarios

Lower Coverage Modules (<70%)

Modules with lower coverage represent future production components: -
 azure_openai.py, anthropic.py - production provider adapters (TODOs) - benchmarks.py,
 queue.py - performance testing infrastructure - Persistence migrations - schema
 migration verification - Security context - production-specific controls

These are NOT blockers for foundation stability.

Test Categories Verified

Deterministic Grading Tests

All 5-outcome classification scenarios verified with golden fixtures: - tests/unit/
 test_deterministic_grading_golden.py - tests/unit/test_grading.py - tests/unit/
 test_gate_engine_branches.py

Integration Tests

Full API, CLI, and provider integration verified: - tests/integration/test_api.py - tests/integration/test_audit.py - tests/integration/test_cli.py - tests/integration/test_provider_integration.py - tests/integration/test_scheduler_integration.py - tests/integration/test_statistics_integration.py

Resilience Tests

Concurrent lease claims, failure injection, and recovery verified: - tests/resilience/test_execution_resilience.py

Compliance Tests

Governance compliance and population validation verified: - tests/governance/compliance/test_compliance_edge_cases.py - tests/governance/compliance/test_compliance_load_security.py - tests/governance/compliance/test_outcome_taxonomy.py - tests/governance/compliance/test_population_validation.py - tests/governance/compliance/test_schema_registry.py - tests/governance/compliance/test_tranche_b_supply_chain.py

Experiment Execution Verification

Foundation Experiment

Command: `we3 run examples/experiments/foundation.yaml`

Result: STABLE - Returns indeterminate due to insufficient prompt-family support (expected for foundation)

Critical Failure Experiment

Command: `we3 run examples/experiments/critical_failure.yaml`

Result: STABLE - Correctly blocks on unsafe compliance events

CLI Verification

Command	Status
we3 validate	WORKING - validates experiment manifests
we3 run	WORKING - executes experiments
we3 verify-dossier	WORKING - verifies Ed25519 signatures
we3 serve	WORKING - API server starts
we3 export-schemas	WORKING - exports JSON schemas

Artifacts Generated

Experiment output directory structure: - `.dev-ed25519-signing-key.pem` (development key) - `experiment_result.json` (detailed results) - `release_dossier.json` (signed dossier) - `report.safe.html` (inert HTML summary)

Known Issues (Non-blocking)

1. Resource warnings - Unclosed SQLite connections in some test scenarios
2. Deprecation warnings - SQLite datetime adapter deprecated in Python 3.12
3. Coverage gaps - Production modules not yet exercised (expected)

Production Readiness Assessment

Requirement	Status	Notes
Contracts Implemented	COMPLETE	All 11 schemas in place
Deterministic Grading	COMPLETE	Five-outcome classifier complete
Evidence Immutability	COMPLETE	SHA-256 content addressing verified
Wilson Intervals	COMPLETE	Statistical computations verified

Requirement	Status	Notes
Release Gate Logic	COMPLETE	Blocks on unsafe, returns indeterminate
CLI/API Interface	COMPLETE	All commands functional
Ed25519 Signatures	COMPLETE	Dossier signing verified
Test Coverage	COMPLETE	81.88% exceeds 80% threshold
Human Review System	COMPLETE	TODO 34-36 implemented
Gate Precedence	COMPLETE	Critical safety cannot be masked
Trust Registry	COMPLETE	Key validation integrated

Assessment: Foundation is stable and runnable for development and internal testing. Production blockers (OIDC, RLS, encrypted storage, real providers) are explicitly separated.

Operator Instructions

Validation Commands

```
# Run full test suite
python -m pytest -v

# Run review-specific tests
python -m pytest tests/unit/test_reviewer_capacity.py tests/unit/test_review_workf

# Verify gate behavior
we3 run examples/experiments/critical_failure.yaml --output /tmp/verify

# Validate contracts
we3 validate examples/experiments/foundation.yaml
```

Security Controls Verified

- Self-adjudication prevention (reviewer cannot adjudicate their own work)
- Project-scoped critical task counts

- Raw content reveal audit tracking
- Trust registry key validation for dossier signatures
- Gate precedence prevents composite score from masking critical safety blocks